

Two-Digit COUNTER

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A two-digit counter is an electronic device that counts from 00 to 99, displaying the count on two 7-segment displays.

With each clock pulse, the displayed value increments, resetting to 00 after reaching 99. The first

decade counter controls the units digit, while the second manages the tens. This configuration is commonly used in digital clocks, event counters, and educational models to illustrate core digital electronics concepts.

Fig. 1 shows the author's prototype.

Circuit and working

The device is built around an LM555 timer (IC1), two decade counter ICs (CD4033: IC2 and IC3), two 7-segment LTS543 displays (DIS1 and DIS2), and a few other components. This two-digit counter effectively counts in decimal format (from 00 to 99).

The LM555 timer (IC1) is configured as a monostable multi-vibrator that generates a pulse. This monostable output serves as the clock input for CD4033 (IC2). IC2 increments on every positive clock edge, providing an output to the

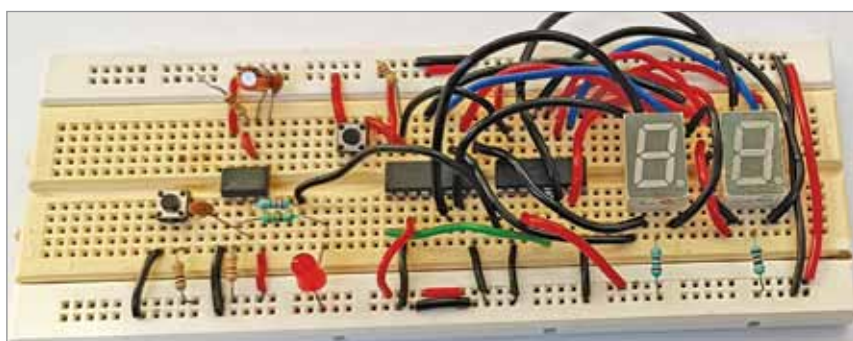


Fig. 1: Author's prototype on a breadboard

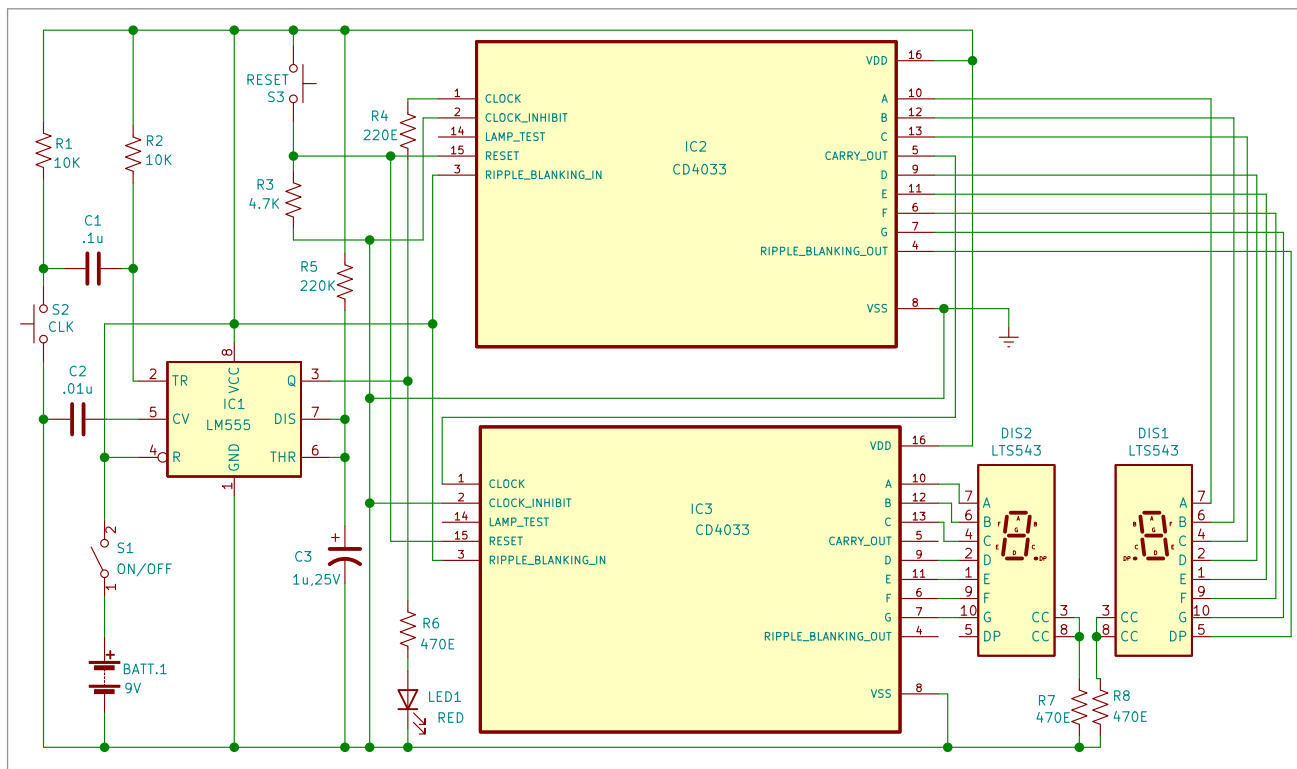


Fig. 2: Circuit diagram